

Technical Design: Offline Synchronization Strategy

1. Architecture Overview

Agri-Sync is engineered for environments with zero-to-intermittent connectivity. The system implements an **Offline-First** model where the local device acts as the primary data source for all mutations. Synchronization with the central Laravel API is treated as a background process, ensuring field technicians face no latency or service interruptions during field operations.

Core Stack

- **Local Persistence:** Browser-based IndexedDB.
 - **Sync Logic:** Workbox + Service Worker (PWA Background Sync API).
 - **Consistency Model:** Optimistic UI with Last-Write-Wins (LWW) conflict resolution.
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2. Local-First Write Strategy

All data entry forms (Irrigation, Fertilization, Phytosanitary, Harvest) bypass the network layer on initial submission.

1. **Validation & Capture:** Input is validated against local business rules.
 2. **IndexedDB Persistence:** The record is serialized and saved to a persistent **Outbox** in IndexedDB.
 3. **Client-Side ID:** Each transaction is assigned a locally generated UUID (`client_id`). This serves as the source of truth for idempotency checks during server-side processing.
 4. **Optimistic Acknowledge:** The user receives an immediate success notification. The internal sync state is set to `pending` .
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3. Background Synchronization (Workbox)

The synchronization engine manages reliable delivery using the **Workbox Background Sync** library.

Automated Replay Engine

The Service Worker intercepts all POST requests directed to `/api/v1/operations/*` . If a network request fails (Offline, Timeout, 5xx Server Error):

- **Queueing:** Workbox stores the failed request in its internal sync queue.
- **Browser-Level Sync:** The browser triggers a `sync` event when it detects a stable connection.
- **Ordering:** Requests are replayed in chronological order to maintain data integrity across related operations.

Manual Synchronization Flow

Technicians can monitor synchronization health via a dedicated "Sync Centre" screen.

- **States:** `pending` (waiting for network), `syncing` (in-flight), `error` (server-side validation failure).
- **Manual Trigger:** A "Sync Now" button forces an immediate replay of the Workbox queue, bypassing the standard browser sync wait intervals.

4. Conflict Resolution & Idempotency

Agri-Sync avoids complex version-branching by utilizing the **Postings** intermediate table pattern.

The Posting Lifecycle

Server-side processing involves two distinct phases:

1. **Ingestion:** Incoming POSTs land in the `postings` table. The `client_id` ensures that even if a Service Worker retries a successful request, the database rejects the duplicate transaction.
2. **Migration:** A background process (or immediate middleware) maps the `payload` from the `postings` table to the specific operation tables.

Conflict Handling

Conflicts are resolved via **Last-Write-Wins (LWW)** using the `updated_at` timestamp. In the context of farm operations, chronological sequence is the primary driver of truth; the server assumes the most recent submission reflects the final intent of the technician.

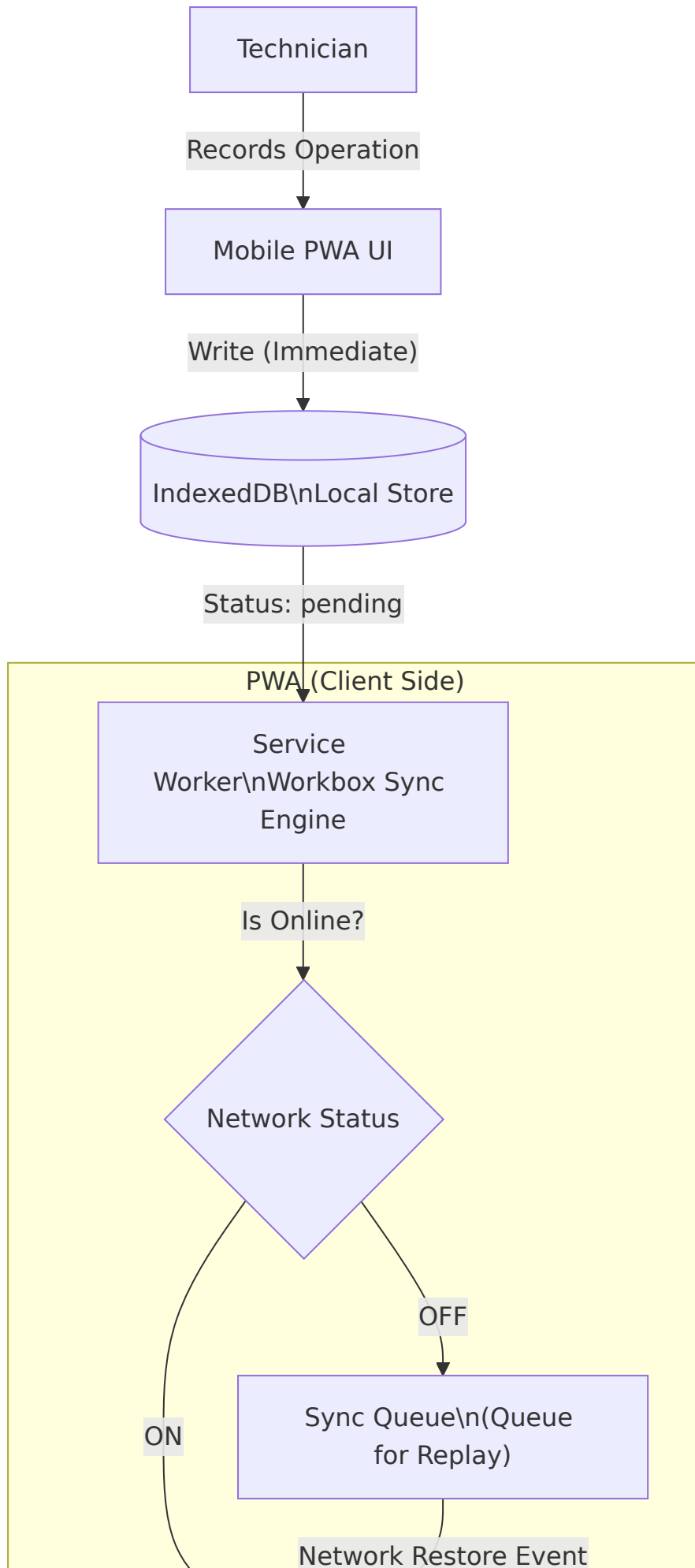
5. Reference Data Hydration

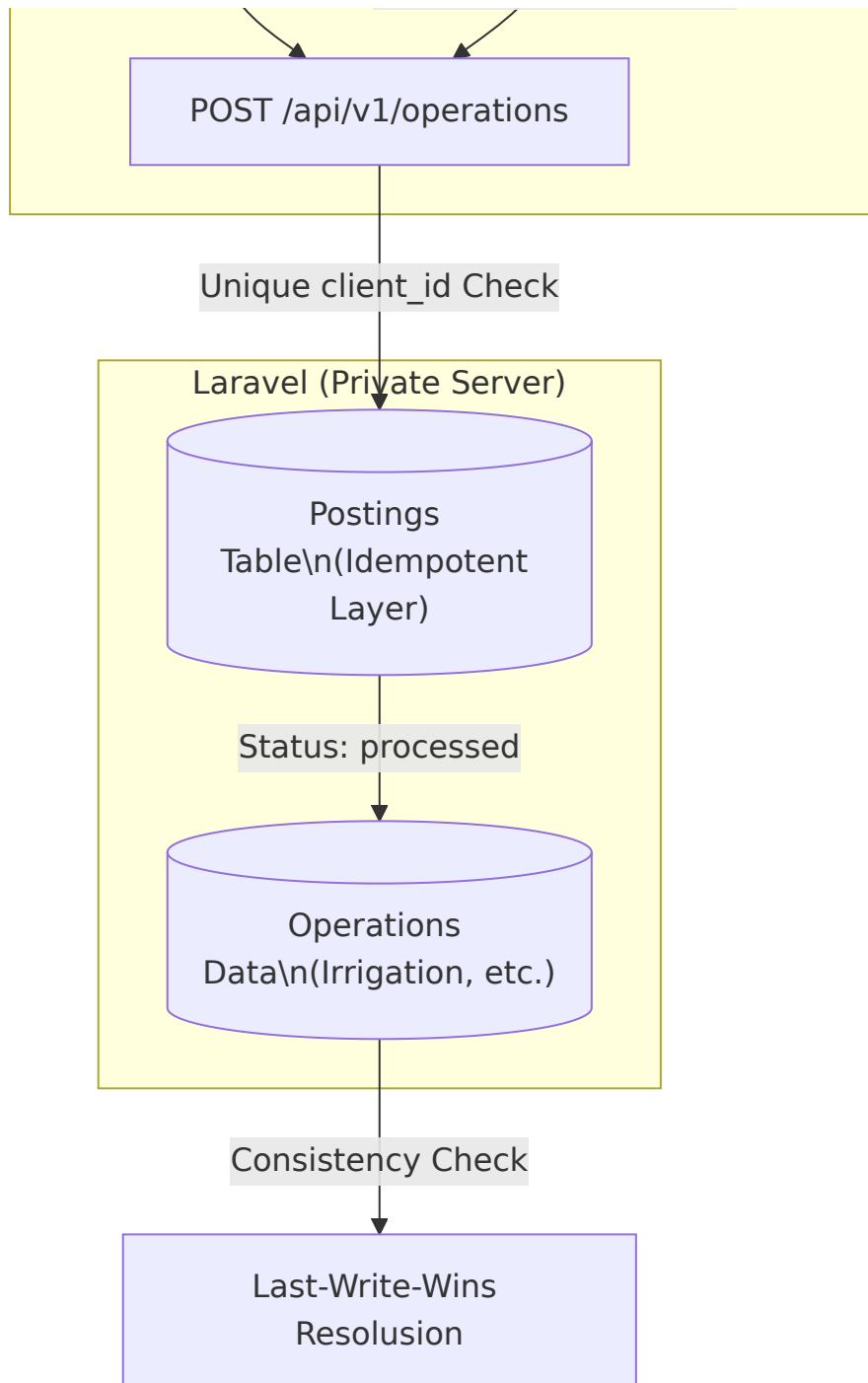
To ensure forms remain functional without an internet connection, reference data is cached locally during the active session.

Resource	Source Endpoint	Refresh Trigger
Plots	<code>GET /api/v1/plots</code>	Login & Pull-to-Refresh
Fertilizers	<code>GET /api/v1/fertilizers</code>	Login & Setting Update
Pesticides	<code>GET /api/v1/pesticides</code>	Login & Setting Update

Reference data is stored as JSON snapshots in a dedicated "Configuration" store within IndexedDB.

6. System Architecture Diagram





7. Compliance & Standards

- **PWA Standards:** Complies with Service Worker Lifecycle management (SW Pre-caching).
- **Persistence:** IndexedDB data persists across browser sessions until explicitly cleared or uninstalled.
- **Data Security:** All synchronization occurs over HTTPS; sensitive payloads are validated server-side before migration from the `postings` table.

